

VIRTUAL UNIVERSITY OF PAKISTAN

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CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

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WhatsApp: 03087122922



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Best of luck!



Question No : 1 of 52

Marks: 1 (Budgeted Time 1 Min)

The conversion of Cauchy Euler equation $4x^2 \frac{d^2y}{dx^2} + 8x \frac{dy}{dx} + y = 0$ after putting $x = e^t$ becomes

Answer (Please select your correct option)

☐ $(4\Delta^2 - 4\Delta - 1)y = 0$

☐ $(4\Delta^2 + 4\Delta + 1)y = 0$

☐ $(4\Delta^2 + 8\Delta + 1)y = 0$

☐ None of them



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Question No : 2 of 52

Marks: 1 (Budgeted Time 1 Min)

Consider a power series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (x-1)^n$ then power series clearly converges for the value of x lies in

Answer (Please select your correct option)

☐ $0 < x < 1$

☒ $0 < x < 2$

☐ $0 < x < 3$

☐ None of them



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Question No : 3 of 52

Marks: 1 (Budgeted Time 1 Min)

Consider a power series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} x^n$ then power series gives an inconclusive result if

Answer (Please select your correct option)

☐ $|x| \leq 1$

☐ $|x| = 1$

☐ $|x| > 1$

☐ None of them



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Question No : 4 of 52

Marks: 1 (Budgeted Time 1 Min)

Suppose that a power series $\sum_{n=1}^{\infty} a_n (x-a)^n$ is represented by a function "f" whose domain is the interval of the convergence of the power series. That function "f" is continuous, differentiable and integrable on

Answer (Please select your correct option)

☐ (a + R, a - R)

☐ (R - a, R + a)

☐ (a - R, a + R)

☐ None of them



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Question No : 5 of 52

Marks: 1 (Budgeted Time 1 Min)

Solution of the D.Equation $4y'' + y = 0$ is

Answer (Please select your correct option)

☐ $y(x) = c_1 \cos \frac{x}{2} + c_2 \sin \frac{x}{2}$



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☐ $y(x) = c_1 \cos \frac{x}{2}$

☐ $y(x) = c_1 \sin \frac{x}{2}$

☐ None of them

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Question No : 6 of 52

Marks: 1 (Budgeted Time 1 Min)

A function f is said to be convergent at a point a if it can be represented by the power series in $(x-a)$ which has

Answer (Please select your correct option)

☐ Divergent series

☐ Convergent series

☐ Both of the 1st and 2nd option

☐ None of them

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Question No : 7 of 52

Marks: 1 (Budgeted Time 1 Min)

If $E(t)=0, R \neq 0$ ($E(t)$ is the source voltage & R is the resistance) then electric vibration of the circuit is said to be

Answer (Please select your correct option)

- ☐ Free damped oscillation
- ☐ Free un-damped oscillation
- ☐ Both damped and un-damped oscillation
- ☐ None of them



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Question No : 8 of 52

Marks: 1 (Budgeted Time 1 Min)

If $E(t)=0, R=0$ ($E(t)$ is the source voltage & R is the resistance) then electric vibration of the circuit is said to be

Answer (Please select your correct option)

- ☐ Free damped oscillation
- ☐ Free un-damped oscillation
- ☐ Both damped and un-damped oscillation
- ☐ None of them

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Question No : 9 of 52

Marks: 1 (Budgeted Time 1 Min)

The Quasi-frequency of the solution $\mathbf{x(t)}$ of free damped motion is given by the number

Answer (Please select your correct option)

☐ $\frac{\sqrt{\omega^2 - \lambda^2}}{2\pi}$



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☐ $\frac{2\pi}{\sqrt{\omega^2 - \lambda^2}}$

☐ $\sqrt{\omega^2 - \lambda^2}$

☐ None of them

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Question No : 10 of 52

Marks: 1 (Budgeted Time 1 Min)

Since $x(t) = \frac{2\sqrt{10}}{3} e^{-t} \sin[3t + 4.391]$ is the solution of $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x = 0$ with $x(0) = -2$, $x'(0) = 0$. So the phase angle is

Answer (Please select your correct option)

☐ 4.193

☒ 4.391

☐ 4.931

☐ None of them



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Question No : 11 of 52

Marks: 1 (Budgeted Time 1 Min)

The standard unit for measurement of inductance is

Answer (Please select your correct option)

☐ Volt

☐ Ohms

☐ Henry

☐ None of them



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Question No : 12 of 52

Marks: 1 (Budgeted Time 1 Min)

The second order linear differential equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 49)y = 0$ is a Bessel equation of degree

Answer (Please select your correct option)

☐ 5

☐ 6

☐ 7

☐ None of them



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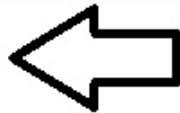
Question No : 13 of 52

Marks: 1 (Budgeted Time 1 Min)

$$J_{-\frac{2}{3}}(x) - J_{\frac{4}{3}}(x) =$$

Answer (Please select your correct option)

☐ $2J'_{\frac{1}{3}}(x)$



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☐ $2J'_{\frac{2}{3}}(x)$

☐ $2J'_{\frac{4}{3}}(x)$

☐ None of them

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Question No : 14 of 52

Marks: 1 (Budgeted Time 1 Min)

Which of the rule in matrices under multiplication does not hold true?

Answer (Please select your correct option)

☐ Commutative law



☐ Associative law

☐ Identity law

☐ None of them

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Question No : 15 of 52

Marks: 1 (Budgeted Time 1 Min)

If a matrix has 3 rows and 2 columns then the given matrix is called

Answer (Please select your correct option)

☐ Column matrix

☐ Square Matrix

☐ Inverse matrix

☐ Rectangular matrix



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Question No : 16 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{bmatrix}$ & $B = \begin{bmatrix} x & y & z & a \\ p & q & r & b \\ l & m & n & o \end{bmatrix}$ then the order of matrix $A \times B$ is

Answer (Please select your correct option)

☐ 2×4



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☐ 2×3

☐ 3×3

☐ None of them

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Question No : 17 of 52

Marks: 1 (Budgeted Time 1 Min)

The given system without the use of matrices $\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} \sin t$ is

Answer (Please select your correct option)

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin 2t$; $\frac{dy}{dt} = x + y + 8 \cos 2t$

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin t$; $\frac{dy}{dt} = x + y + 8 \cos t$

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin t$; $\frac{dy}{dt} = x + y + 8 \sin t$

☐ None of them

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Marks: 1 (Budgeted Time 1 Min)

The given system without the use of matrices $\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} e^{-t}$ is

Answer (Please select your correct option)

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin 2t$; $\frac{dy}{dt} = x + y + 8 \cos 2t$

☐ $\frac{dx}{dt} = 3x - 7y + 4e^{-t}$; $\frac{dy}{dt} = x + y + 8e^{-t}$

☐ $\frac{dx}{dt} = 3x - 7y + 4e^t$; $\frac{dy}{dt} = x + y + 8e^{-t}$

☐ None of them



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Question No : 19 of 52

Marks: 1 (Budgeted Time 1 Min)

The coefficient matrix of the following homogeneous system of differential equation $\frac{dx}{dt} = 3x + 2y$, $\frac{dy}{dt} = x + 2y$ is

Answer (Please select your correct option)

☐ $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$

☐ None of them



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Question No : 20 of 52

Marks: 1 (Budgeted Time 1 Min)

$$\begin{vmatrix} 4-\lambda & 1 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix} = 0 \text{ gives}$$

Answer (Please select your correct option)

☐ $\lambda = 4$ of multiplicity of 1

☐ $\lambda = 4$ of multiplicity of 2

☐ $\lambda = 4$ of multiplicity of 3

☐ None of them



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Question No : 21 of 52

Marks: 1 (Budgeted Time 1 Min)

By applying the Operator method or systematic elimination on a system of linear homogeneous or linear non-homogeneous differential equations we always get a

Answer (Please select your correct option)

☐ Single linear differential equation



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☐ Double linear differential equation

☐ Partial linear differential equation

☐ None of them

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Question No : 22 of 52

Marks: 1 (Budgeted Time 1 Min)

For the system of differential equations $\frac{dy}{dt} = 2x$, $\frac{dx}{dt} = 3y$ the independent variable(s) is (are)

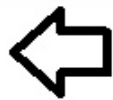
Answer (Please select your correct option)

☐ x, t

☐ y, t

☐ x, y

☐ t



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Question No : 23 of 52

Marks: 1 (Budgeted Time 1 Min)

The differential equation $2\frac{dy}{dx} + x^2y = 2x + 3, y(0) = 5$ is

Answer (Please select your correct option)

Linear

☐

Nonlinear

☒

Linear with fixed constants

☐

Undeterminable to be linear or nonlinear

☐



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Question No : 24 of 52

Marks: 1 (Budgeted Time 1 Min)

If A is a square matrix and its determinant is zero, then

Answer (Please select your correct option)

☐ A is singular matrix.



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☐ A is non singular matrix.

☐ A is scalar matrix.

☐ A is diagonal matrix.

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Question No : 25 of 52

Marks: 1 (Budgeted Time 1 Min)

The Period of oscillator in the solution $X=50\sin(20t+8.5)$ is

Answer (Please select your correct option)

☐ 0.17643

☐ 0.32045

☒ 0.31400

☐ 0.58000

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Question No : 26 of 52

Marks: 1 (Budgeted Time 1 Min)

An electronic component of an electronic circuit that has the ability to store charge and opposes any change of voltage in the circuit is called

Answer (Please select your correct option)

☐ Inductor

☐ Resistor

☐ Capacitor

☐ None of them



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Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

If determinant $\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix} = 0$, then

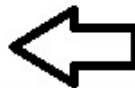
Answer (Please select your correct option)

☐ $D^2 - 4D + 1 = 0$

☐ $D^2 - 4D + 3 = 0$

☐ $D^2 - 4D + 4 = 0$

☐ None of them



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Question No : 28 of 52

Marks: 1 (Budgeted Time 1 Min)

A rectangular arrangement of numbers or functions enclosed in the square brackets is called

Answer (Please select your correct option)

- ☐ Equation
- ☐ derterminant
- ☒ Matrix
- ☐ None of them



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Question No : 29 of 52

Marks: 1 (Budgeted Time 1 Min)

If wroskian of the solution vectors X_1 & X_2 is zero, then vectors are

Answer (Please select your correct option)

☐ Linearly Independent

☒ Linearly dependent

☐ None of them

☐ Parallel

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Question No : 30 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, then eigen values are

Answer (Please select your correct option)

☐ 1,2

☐ 0,1

☐ 0,2

☐ None of them



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Question No : 31 of 52

Marks: 1 (Budgeted Time 1 Min)

Let λ be an eigen value of a non zero square matrix A. Then the equation $\det(A - \lambda I) = 0$ is called

Answer (Please select your correct option)

☐ Trivial equation

☒ Characteristics equation

☐ Non-trivial equation

☐ None of them



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but check the topic
of non-trivial
equation

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Question No : 32 of 52

Marks: 1 (Budgeted Time 1 Min)

Eigen values of the following homogeneous system of Differential equation $\frac{dx}{dt} = x$, $\frac{dy}{dt} = 2x + 2y$ with coefficient matrix $\begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$

Answer (Please select your correct option)

☐ $\lambda = 2, 2$

☐ $\lambda = 1, 1$

☐ None of them

☒ $\lambda = 1, 2$



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Question No : 33 of 52

Marks: 1 (Budgeted Time 1 Min)

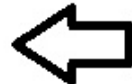
The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an

Answer (Please select your correct option)

☐ partial differential equation.

but in think 1 ,,

☒ ordinary differential equation.



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☐ polynomial equation.

☐ transcendental equation.

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Question No : 34 of 52

Marks: 1 (Budgeted Time 1 Min)

The differential equation $dx + \left(\frac{x}{y} - \sin y\right) dy = 0$ is -----

Answer (Please select your correct option)

☐ Exact.

☐ Non-exact.

☐ Homogenous.

☐ Linear.



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Question No : 35 of 52

Marks: 1 (Budgeted Time 1 Min)

If $f(x)$ and $g(x)$ are linearly dependent on I , then-----.

Answer (Please select your correct option)

- ☐ $W(f,g)(x) \neq 0$ (Wronskian) for all x in the interval I .
- ☐ $W(f,g)(x) = 0$ (Wronskian) for all x in the interval I .
- ☐ $W(f,g)(x)$ may or may not be zero for all x in the interval I .
- ☐ $W(f,g)(x)$ is not defined for all x in the interval I .



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Question No : 36 of 52

Marks: 1 (Budgeted Time 1 Min)

If the auxiliary equation has roots $-2, -2$, then the general solution of the differential equation is _____

Answer (Please select your correct option)

☐ $y = c_1 e^{-2x} + c_2 e^{-2x}$

☐ $y = c_1 e^{-2+2x}$

☐ $y = c_1 e^{2x} + c_2 e^{-2x}$

☐ $y = (c_1 + c_2 x) e^{-2x}$



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Question No : 37 of 52

Marks: 1 (Budgeted Time 1 Min)

If the annihilator operator of $2x$ is D^2 and of xe^{3x} is $(D - 3)^2$, then which of the following is the correct option?

Answer (Please select your correct option)

- ☐ The annihilator operator of $2x + xe^{3x}$ is $(D - 3)^2$.
- ☐ The annihilator operator of $2x + xe^{3x}$ is D^2 .
- ☐ The annihilator operator of $2x + xe^{3x}$ is $D^2 + (D - 3)^2$.
- ☐ The annihilator operator of $2x + xe^{3x}$ is $D^2 (D - 3)^2$.



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Question No : 38 of 52

Marks: 1 (Budgeted Time 1 Min)

If $y_1 = x$ and $y_2 = xe^x$ are the first and second solution of $x^2 \frac{d^2 y}{dx^2} - x(x+2) \frac{dy}{dx} + (x+2)y = 0$ on $(0, \infty)$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- ☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly dependent.
- ☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly independent.
- ☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly independent.
- ☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly dependent.



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Question No : 39 of 52

Marks: 1 (Budgeted Time 1 Min)

If $y_1 = xe^{-x}$ is the first solution of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$, then which of the following is true for it?

Answer (Please select your correct option)

- ☐ Its second solution is $xe^{-x} \int \frac{2}{e^{-2x}} dx$
- ☐ Its second solution is $xe^{-x} \int \frac{2}{x^2 e^{-2x}} dx$
- ☐ Its second solution is $xe^{-x} \int \frac{1}{x^2 e^{-2x}} dx$
- ☐ Its second solution is $xe^{-x} \int \frac{1}{x^2} dx$



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Question No : 40 of 52

Marks: 1 (Budgeted Time 1 Min)

If $x^2 \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = 0$ is the particular form of $\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ $P(x) = \frac{2}{x^2}$ and $Q(x) = 0$

☐ $I.F = e^x$ $P(x) = -2$ and $Q(x) = x^2$

☐ $P(x) = \frac{-2}{x^2}$ and $Q(x) = 0$

☐ $P(x) = 0$ and $Q(x) = \frac{-2}{x^2}$

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For differential equation $x(x-1)y'' + (3x-1)y' + y = 0$ if $y_1 = \sum_{k=0}^{\infty} x_k = 1 + x + x^2 + \dots = \frac{1}{1-x}$; $|x| < 1$ is one solution about regular singularity $x = 0$, then which method will be use to find 2nd solution?

Answer (Please [click here](#) to Add Answer)

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frobenius, i think and .

$$y_2 = y_1(x) \int \frac{e^{-\int P dx}}{y_1^2} dx$$

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Question No : 42 of 52

Marks: 2 (Budgeted Time 4 Min)

Discuss the linearly dependence of solution vectors.

Answer (Please [click here](#) to Add Answer)

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Any two solution vectors X_1 and X_2 are linearly dependent if and only if one of the two vectors is a constant multiple of the other.

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What is Legendre's differential equation

An

Lecture 34

Legendre's Differential Equation

A second order linear differential equation of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendre's polynomial of degree n and is denoted by $P_n(x)$.

We assume a solution of the form $y = \sum_{k=0}^{\infty} C_k x_k$

$$\therefore (1-x^2)y'' - 2xy' + n(n+1)y =$$

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If the complementary solution of the following differential equation is $c_1 \sin 2x + c_2 \cos 2x$, then what will be the general form of its particular solution:

$$\frac{d^2 y}{dx^2} + 4y = \cos 2x$$

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Answer (Please [click here](#) to Add Answer)



$$y_p = A \sin 2x + B \cos 2x$$

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Question No : 45 of 52

Marks: 3 (Budgeted Time 6 Min)

Can the power series method be directly applied if the coefficients $P(x)$ and $Q(x)$ for the differential equations $y'' + P(x)y' + Q(x)y = 0$ are not polynomials?
If $P(x) = \sin x$ and $Q(x) = \cos x$, then how we solve it?

Answer (Please [click here](#) to Add Answer)



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Write the homogenous system of differential equations

$$2 \frac{dx}{dt} - 5x + \frac{dy}{dt} = 5e^t$$

$$\frac{dx}{dt} - x + \frac{dy}{dt} = e^t$$

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Answer (Please [click here](#) to Add Answer)



$$(2D - 5)x + Dy = 5e^t$$

$$(D - 1)x + Dy = e^t$$

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Question No : 47 of 52

Marks: 3 (Budgeted Time 6 Min)

Find the characteristic equation of coefficient matrix of the following system

$$\frac{dx}{dt} = -3x - 2y$$

$$\frac{dy}{dt} = 5x + 7y$$

Answer (Please [click here](#) to Add Answer)

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$$\begin{vmatrix} -3-\lambda & -2 \\ 5 & 7-\lambda \end{vmatrix} = -21 + 3\lambda - 7\lambda + \lambda^2 + 10$$
$$\lambda^2 - 4\lambda - 11 = 0$$
$$\lambda = \frac{4 \pm \sqrt{16 - 4(-11)}}{2} = \frac{4 \pm \sqrt{60}}{2}$$

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Determine whether the following functions are linearly dependent or linearly independent ?

$$y_1 = 9 \cos(2x), y_2 = 2 \cos^2 x - 2 \sin^2 x$$

Answer (Please [click here](#) to Add Answer)

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$$\begin{vmatrix} 9 \cos 2x & 2(\cos^2 x - \sin^2 x) \\ -18 \sin 2x & 2(2 \cos x(-\sin x) - 2 \sin x(\cos x)) \end{vmatrix}$$

$$\begin{vmatrix} 9 \cos 2x & 2(\cos^2 x - \sin^2 x) \\ -18 \sin 2x & -4 \cos x(\sin x) - 4 \sin x(\cos x) \end{vmatrix}$$

$$\begin{vmatrix} 9 \cos 2x & 2(\cos^2 x - \sin^2 x) \\ -18 \sin 2x & -8 \cos x(\sin x) \end{vmatrix}$$

$$-8 \cos x(\sin x)[9 \cos 2x] + 18 \sin 2x[2(\cos^2 x - \sin^2 x)]$$

independent

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Question No : 49 of 52

Marks: 5 (Budgeted Time 10 Min)

Find the general solution of the given differential equation on $(0, \infty)$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left(x^2 - \frac{1}{64}\right)y = 0$$

Answer (Please [click here](#) to Add Answer)

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$$v^2 = \frac{1}{64}$$

$$v = \pm \frac{1}{8}$$

$$y = C_1 j_{1/8}(x) + C_2 j_{-1/8}(x)$$

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Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-t}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer (Please [click here](#) to Add Answer)

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$$w = \begin{vmatrix} e^{-3t} & e^{-t} & e^{6t} \\ -3e^{-3t} & -e^{-t} & 6e^{6t} \\ 9e^{-3t} & e^{-t} & 36e^{6t} \end{vmatrix}$$

also find w1, w2, w3

$$w_2 = \begin{vmatrix} e^{-3t} & 0 & e^{6t} \\ -3e^{-3t} & 0 & 6e^{6t} \\ 9e^{-3t} & 3 + 4e^{-t} & 36e^{6t} \end{vmatrix}$$

$$w_1 = \begin{vmatrix} 0 & e^{-t} & e^{6t} \\ 0 & -e^{-t} & 6e^{6t} \\ 3 + 4e^{-t} & e^{-t} & 36e^{6t} \end{vmatrix}$$

$$w_3 = \begin{vmatrix} e^{-3t} & e^{-t} & 0 \\ -3e^{-3t} & -e^{-t} & 0 \\ 9e^{-3t} & 3 + 4e^{-t} & 0 \end{vmatrix}$$

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Write the following system in matrix form

$$\frac{dx}{dt} = -3x + 4y + e^{-t} \sin 2t$$

$$\frac{dy}{dt} = 5x + 9y + 4e^{-t} \cos 2t$$

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Answer (Please [click here](#) to Add Answer)

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$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix} = \begin{bmatrix} -3 & 4 \\ 5 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} \sin 2t \\ 4 \cos 2t \end{bmatrix} e^{-t}$$

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Question No : 52 of 52

Marks: 5 (Budgeted Time 10 Min)

As $x=0$ is regular singular point of the differential equation $10xy'' + 10y' - y = 0$, find the relation of C_{k+1} and C_k in the process of finding solution.

Answer (Please [click here](#) to Add Answer)

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